

AMENDMENT NO. 1 JUNE 2020

TO

IS 17216 : 2019 POLYETHYLENE MULCH FILMS FOR AGRICULTURE AND HORTICULTURE — SPECIFICATION

(Page 1, clause 2) — Insert the following at the end:

<i>IS No./Other publication</i>	<i>Title</i>
ISO 9845-1 : 1992	Solar energy — Reference solar spectral irradiance at the ground at different receiving conditions Part 1: Direct normal and hemispherical solar irradiance for air mass 1,5
ISO 9060 : 1990	Solar energy — Specification and classification of instruments for measuring hemispherical solar and direct solar radiation

(Page 2, clause 5.1.2) — Substitute the following for the existing clause:

‘5.1.2 Melt Flow Index

The melt flow index of polyethylene resin from which mulch film is made shall be between 0.1 to 2.5 g/10 min when determined in accordance with IS 13360 (Part 4/Sec 1) under test temperature of 190°C and test load of 2.16 kg.’

(Page 3, clause 5.5) — Insert the following new clause:

‘5.6 Solar Reflectance for Reflective Mulch Film

When tested as per Annex E, the solar reflectance of the reflective mulch film shall be as given below:

- Silver Black mulch film, 32 percent, *Min*; and
- White Black mulch film 60 percent, *Min*’

(Page 7, Annex D) — Insert the following new ‘Annex E’ and renumber the subsequent annexes:

‘ANNEX E

(Clause 5.6)

DETERMINATION OF SOLAR REFLECTANCE

E-1 PRINCIPLE

The reflectance spectra of a film are acquired by using a UV-VIS-NIR spectrophotometer equipped with an integrating sphere. The solar reflectance, R_s , is calculated according to the method described in ISO 9845-1.

E- 2 TERMS AND DEFINITIONS

For the purposes of this annex, the following definitions apply.

E-2.1 Air Mass Zero (AM0) — Solar radiation quantities outside the earth’s atmosphere at the mean earth sun distance.

E-2.2 Air Mass (AM) — Ratio of the mass of atmosphere in the actual observer — sun path to the mass that would exist if the observer were at sea level, at standard barometric pressure, and the sun were directly overhead.

NOTE — Air mass varies with the elevation of the sun and the local barometric pressure, which changes with altitude for a sun zenith angle, Z , of 62° or less, and local atmospheric pressure, P , where P_0 is standard atmospheric pressure, $AM = P/(P_0 \cos Z)$.

E-2.3 Direct Solar Irradiance — On a given plane receiver surface, the ratio of the radiant fluxes received from a small solid angle centred on the sun’s disk to the area of that surface (unit: watts per square metre, Wm^{-2}).

NOTE — If the plane is perpendicular to the axis of the solid angle, the direct normal solar irradiance is received. For appropriate radiometers of modern design, the applied solid angles correspond to field-of-view angles of less than 6° .

E-2.4 Hemispherical Solar Irradiance — On a given plane, the ratio of the solar radiant flux received from the sky hemisphere above — including the direct solar radiant flux to the area of the plane (unit: watts per square metre, Wm^{-2}).

E-2.5 Spectral Solar Irradiance (E_λ) — Solar irradiance E per unit wavelength interval at a given wavelength λ (unit: watts per square metre per micrometre, $Wm^{-2}\mu m^{-1}$).

$$E_{\lambda} = dE/d\lambda$$

NOTE — The reference spectra are evaluated in the range of 0.305 μm to 4.045 μm . The relevant spectral range of the measurement of the solar radiation is confined to the range from 0.3 μm to 3 μm . This accounts for 98.5 percent of the spectra (further details are given in ISO 9060, Solar energy — Specification and classification of instruments for measuring hemispherical solar and direct solar radiation). However, the spectra in this part of ISO 9845 are confined to the range from 0.305 μm to 4.045 μm .

E- 2.6 Meteorological Optical Range — Horizontal distance V at which the contrast between a black target and the sky above the horizon is equal to the threshold contrast ε_0 ;

$$V = \frac{1}{\sigma} \ln \frac{1}{\varepsilon_0}$$

where σ is the atmospheric extinction coefficient in reciprocal metres, and ε_0 is a parameter equal to 0.05; thus

$$\ln \frac{1}{\varepsilon_0} \approx 3.0$$

E-2.7 Specular, indicates that the flux leaves a surface or medium at an angle that is numerically equal to the angle of incidence, lies in the same plane as the incident ray and the perpendicular, but is on the opposite side of the perpendicular to the surface.

E-3 APPARATUS

E-3.1 General

UV-VIS-NIR spectrophotometer equipped with an integrating sphere, collecting both the specular and the diffusively reflected part of the light, as shown in Fig. 1.

The internal side of integrating sphere is covered with a highly reflecting material. The integrating sphere collects the diffusively reflected light from a surface. Specular part of the light could also be collected but often it can be ignored.

Behind the specimen a radiation trap (a cavity having a black internal surface) is placed in order to absorb the light which is not absorbed or reflected by the specimen.

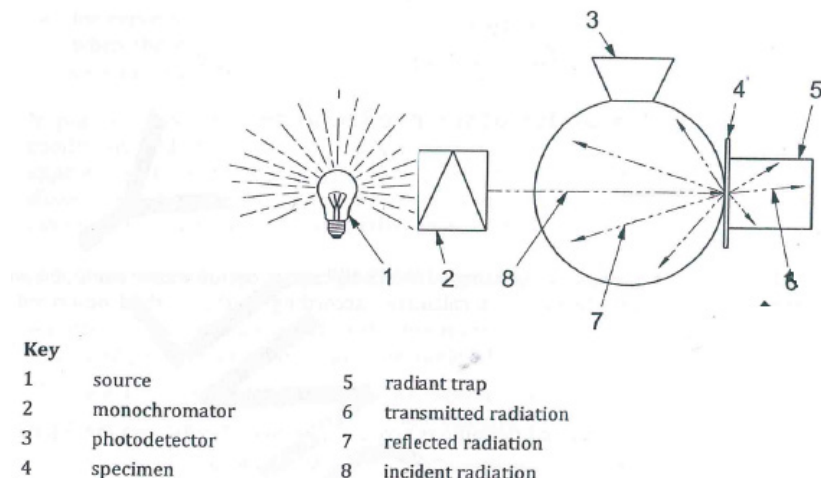


FIG. 1 INTEGRATING SPHERE

E-3.2 Test Specimens

Three specimens shall be cut from the film.

E-3.3 Procedure

The measurements are performed using the UV-VIS-NIR spectrophotometer equipped with the integrating sphere, collecting both the specular and the diffusely reflected part of the light. The test specimen shall be placed at the exit port of the sphere, that is at the rear of the integrating sphere. The inside of the film as seen from the reel shall be directed towards the sphere (*see* Fig. 1).

A reflectance spectrum, $R_{\text{spec}}(\lambda)$ is acquired by the UV-VIS-NIR-spectrophotometer for each specimen between 300 nm and 2500 nm on one layer of the film. The contribution from wavelengths greater than 2 500 nm can be neglected.

E-3.4 Calculation of the Solar Reflectance, R_s

The solar reflectance R_s , defined as the fraction of the total direct solar irradiation which is reflected by a body, shall be calculated according to ISO 9845-1, Formula (1), that is:

$$R_s = \frac{\int_0^{\infty} R(\lambda) E(\lambda) d\lambda}{\int_0^{\infty} E(\lambda) d\lambda}$$

where

$E(\lambda)$ solar spectral irradiance; and

$R(\lambda)$ absolute reflectance.

Reflectance spectra, $R_{\text{spec}}(\lambda)$, the UV-VIS-NIR-spectrophotometer are measured relatively to a reflectance standard with a known absolute reflectance value, $R_{\text{ref}}(\lambda)$.

$R(\lambda)$ is calculated from the reflectance spectrum from the following formula:

$$R(\lambda) = R_{\text{spec}}(\lambda)R_{\text{ref}}(\lambda).$$

where

$R_{\text{spec}}(\lambda)$ reflectance of the specimen; and

$R_{\text{ref}}(\lambda)$ reflectance of a reflectance standard.

Calculate the arithmetic average value of the three measurements.

An example of a reflectance reference¹⁾ is pressed barium sulphate (BaSO_4), $R_{\text{ref}}(\lambda)$ for BaSO_4 can be obtained from National Physical Laboratory. Other reflectance references exist. The manufacturers provide the absolute reflectance values.

¹⁾ Pressed Eastman barium sulphate (BaSO_4) is an example of a suitable product available commercially. This information is given for the convenience of users and does not constitute an endorsement. Equivalent products may be used if they can be shown to lead to the same results.'

(Page 4, clause 7.2.4, line 4) — Substitute 'Coefficient of friction' for 'Slip'.